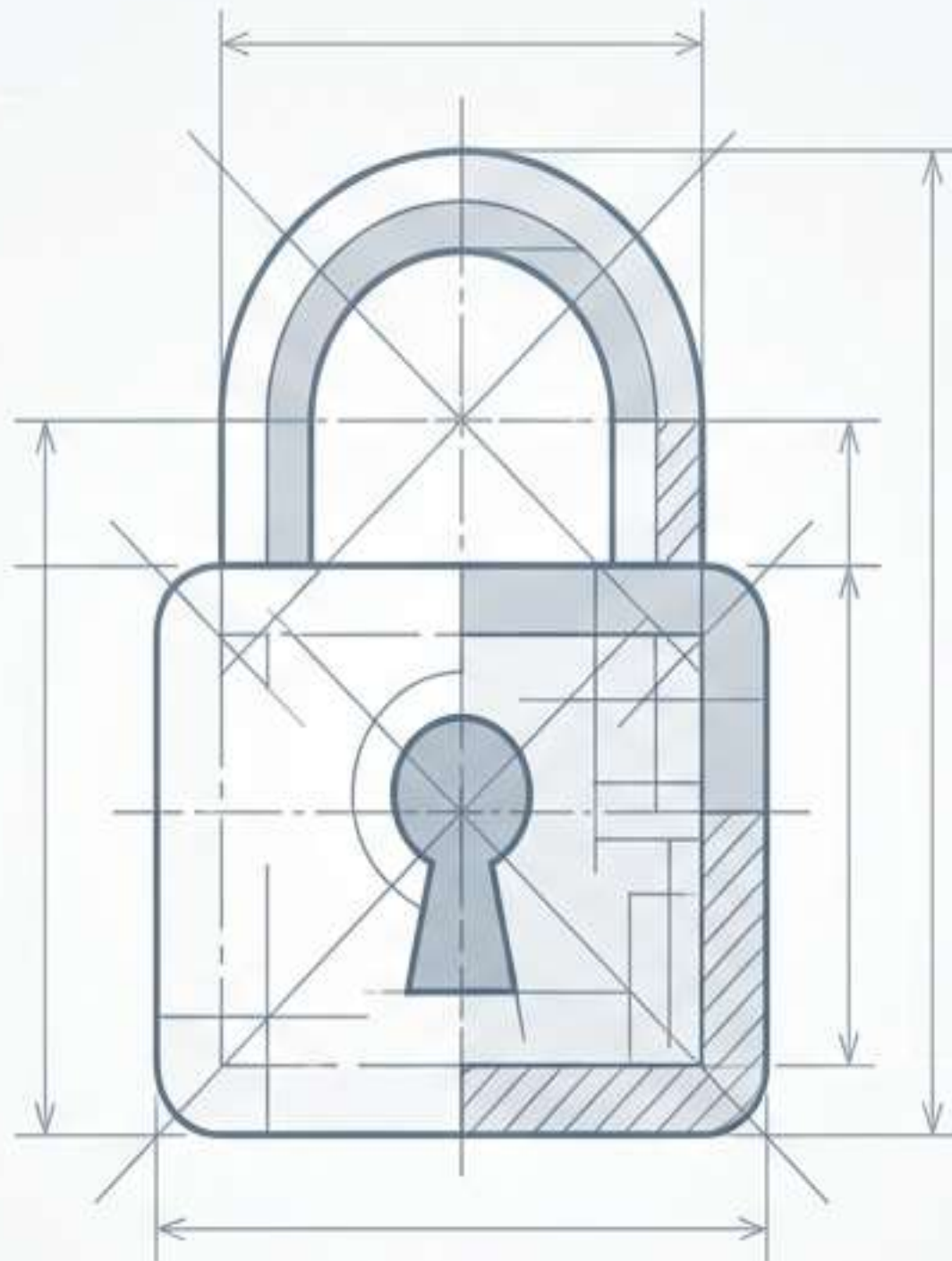




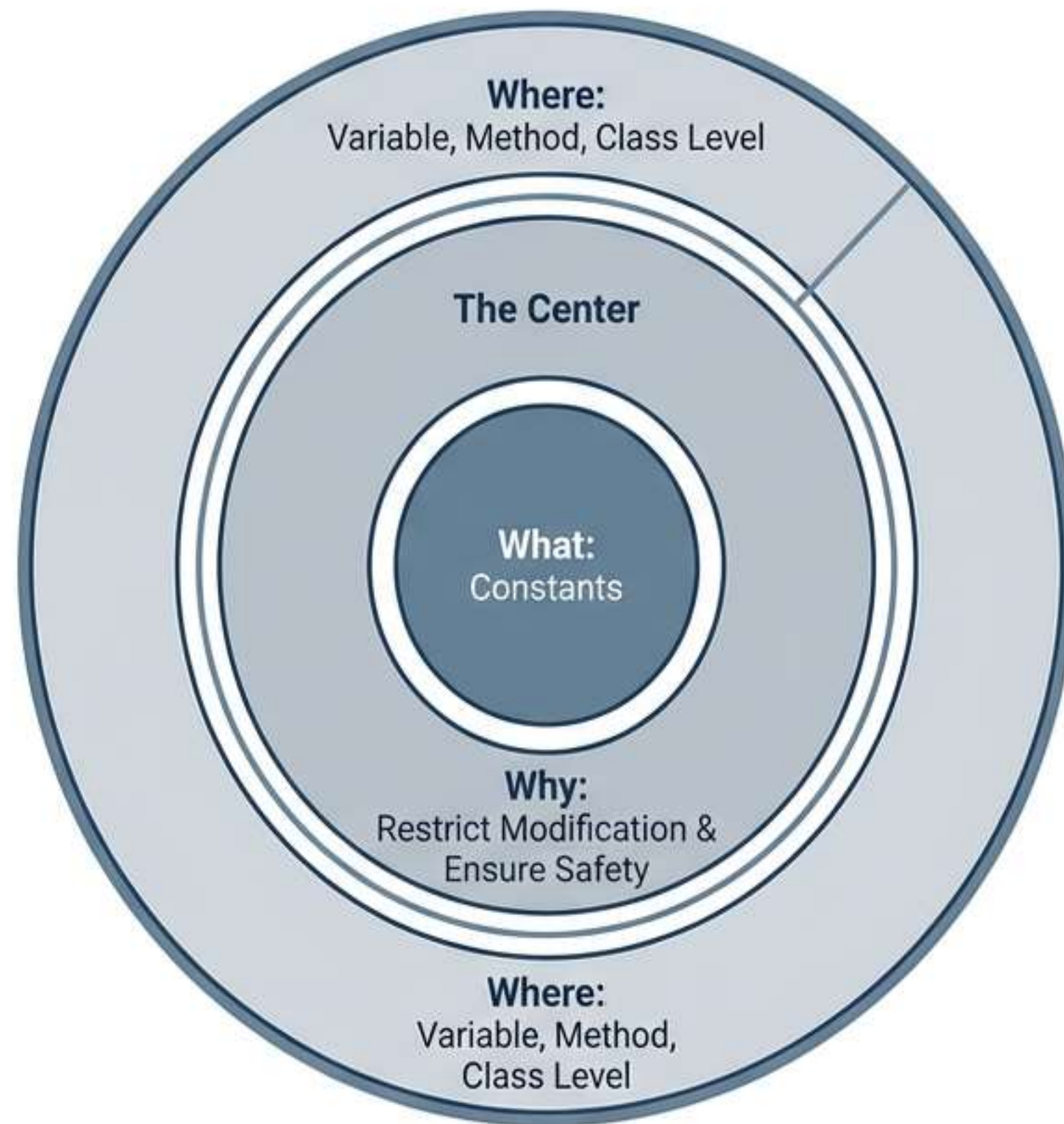
Java Mechanics: The final Keyword

Ensuring Immutability, Security, and Code Integrity

The Power of Restriction

Definition: 'final' is a non-access modifier used to restrict modifications to code elements. It creates a contract that the state or behaviour cannot change.

The Golden Circle of final



Three Pillars of Immutability

Variables



Data Integrity

Acts as a constant. Value cannot be reassigned once initialised.

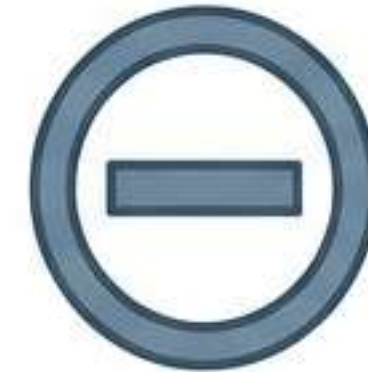
Methods



Behavioural Integrity

Prevents logic alteration. Subclasses cannot override the method implementation.

Classes



Structural Integrity

Prevents inheritance hierarchy. The class cannot be subclassed (e.g., String class).

Syntax and Implementation Details

```
1 // 1. Defining Constants
2 final float pi = 3.14f;
3
4
5
6 // 2. Locking Logic (Methods)
7 public final void methodSample() {
8     // Critical algorithm code
9 }
10
11
12
13
14 // 3. Sealing Types (Classes)
15 public final class Vehicle {
16     // Class body
17 }
18
19
20
```

Variables:

Must be initialised before constructor completes.

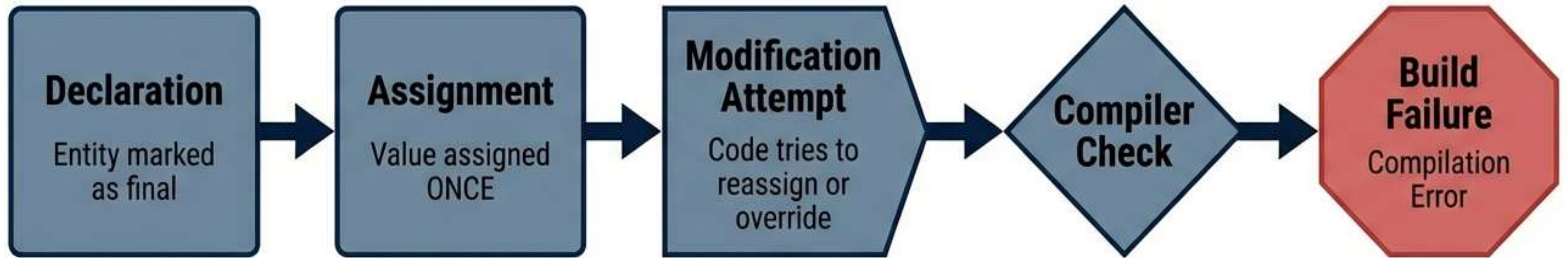
Methods:

Cannot be overridden by child classes.

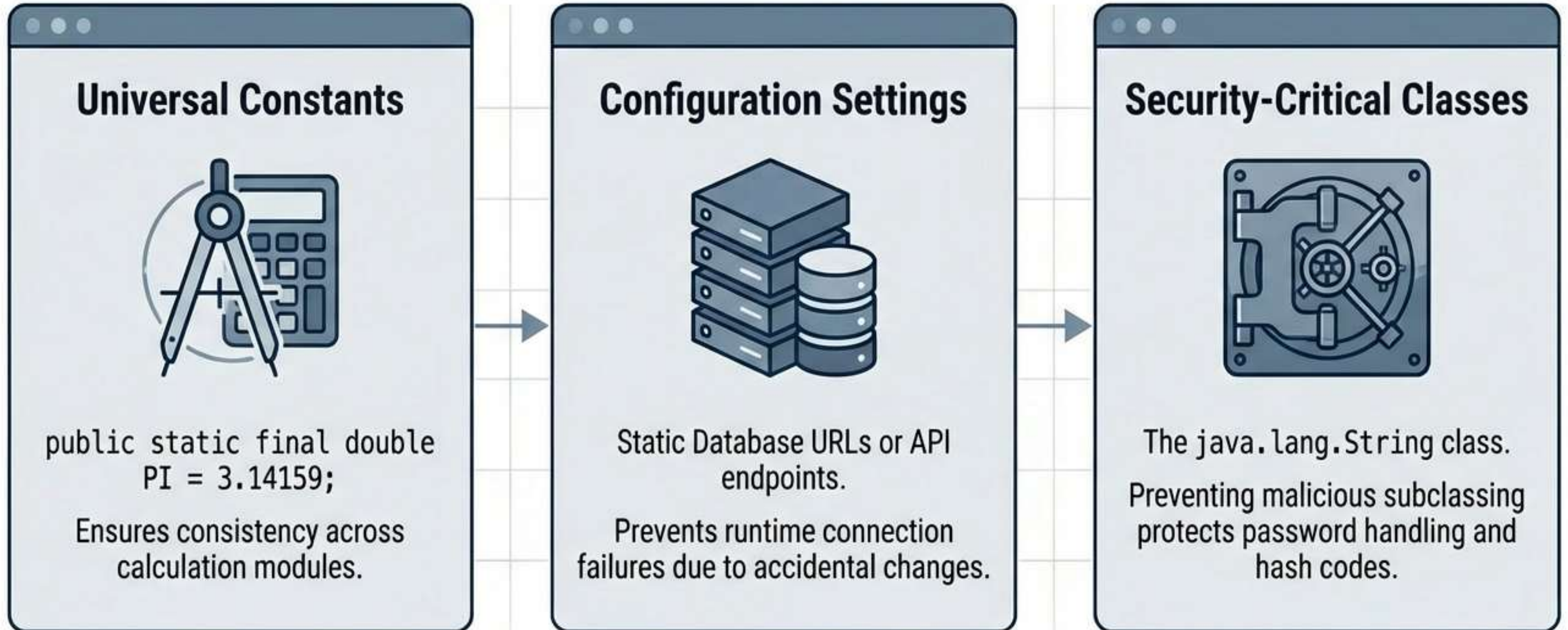
Classes:

Cannot be extended using 'extends' keyword.

Compilation Enforcement Workflow



Application in Enterprise Development



Technical Interview Preparation

Cheat Sheet **Abstract vs. Final**

Can an abstract class be final?

NO. It is a contradiction. Abstract requires extension; Final prevents it.

Cheat Sheet **Final Constructors**

Can a constructor be final?

NO. Constructors are not inherited, so they cannot be overridden.

Cheat Sheet **The 'F' Word Trio**

final vs. finally vs. finalize?

final (Restriction Keyword), finally (Exception Block), finalize (Garbage Collection Method).

Core Competencies Recap



Variable Scope: Locks value assignment.
(Constant, Read-Only).



Method Scope: Locks implementation logic.
(No Overriding, Security).



Class Scope: Locks inheritance hierarchy.
(Immutable, Sealed).

Executive Summary

- The final keyword is a cornerstone of robust Java architecture.
- It transforms mutable elements into reliable constants.
- Correct usage prevents accidental data corruption and malicious code injection.
- Mastery of final is essential for creating secure, enterprise-grade applications.

